

SMALL COUNTEREXAMPLES TO A TRACE CONJECTURE OF FRANKL AND WANG

ABSTRACT. Frankl and Wang conjectured an exact formula for $m(n, \ell + 1, 3 \cdot 2^{\ell-1} + 1)$ for every $n > \ell > 0$. We prove

$$m(6, 5, 25) = 40, \quad m(7, 5, 25) \geq 58, \quad m(7, 6, 49) \geq 80,$$

where their formula gives 37, 55, and 73, respectively. Thus the conjecture fails already at $n = \ell + 2$ for $\ell = 4$ and 5. The first value is determined using the Kruskal–Katona theorem; the other counterexamples are explicit down-sets.

1. INTRODUCTION

For $\mathcal{F} \subseteq 2^{[n]}$ and $Y \subseteq [n]$, let

$$\mathcal{F}|_Y = \{F \cap Y : F \in \mathcal{F}\}.$$

The arrow relation $(n, m) \rightarrow (a, b)$ means that every $\mathcal{F} \subseteq 2^{[n]}$ with $|\mathcal{F}| \geq m$ has an a -set Y satisfying $|\mathcal{F}|_Y| \geq b$. Write

$$m(n, a, b) = \min\{m : (n, m) \rightarrow (a, b)\}.$$

Frankl and Wang conjectured [1, Conjecture 1.3] that, for all $n > \ell > 0$,

$$(1) \quad m(n, \ell + 1, 3 \cdot 2^{\ell-1} + 1) = 1 + \prod_{0 \leq i < \ell} \left\lfloor \frac{n + \ell + i}{\ell} \right\rfloor.$$

The same statement was subsequently recorded as an open problem in [4, Conjecture 2].

Theorem 1. *One has*

$$m(6, 5, 25) = 40, \quad m(7, 5, 25) \geq 58, \quad m(7, 6, 49) \geq 80.$$

Consequently, (1) is false for $(n, \ell) = (6, 4), (7, 4), (7, 5)$.

For these parameters, the right-hand side of (1) is, respectively, 37, 55, and 73. When $n = \ell + 1$, the proposed threshold equals the target trace size, so the assertion is tautological. Thus $(6, 4)$ and $(7, 5)$ are the first nontrivial cases for $\ell = 4$ and 5.

All families constructed below are down-sets. For a down-set $\mathcal{F}$,

$$(2) \quad \mathcal{F}|_Y = \mathcal{F} \cap 2^Y.$$

We write ijk for $\{i, j, k\}$ and $\binom{X}{\leq r} = \{S \subseteq X : |S| \leq r\}$.

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2. THE EXACT VALUE $m(6, 5, 25) = 40$

Let

$$D = \{136, 145, 156, 234, 236, 245\}, \quad A = \{1235, 1246, 3456\},$$

and define

$$(3) \quad \mathcal{F} = \binom{[6]}{\leq 2} \cup \left(\binom{[6]}{3} \setminus D \right) \cup A.$$

No member of D is contained in a member of A , so $\mathcal{F}$ is a down-set. Moreover, D is 3-regular and A is 2-regular on $[6]$, while

$$|\mathcal{F}| = 22 + (20 - 6) + 3 = 39.$$

For $Y_x = [6] \setminus \{x\}$, equation (2) gives

$$|\mathcal{F}|_{Y_x} = \sum_{i=0}^2 \binom{5}{i} + (\binom{5}{3} - (6 - 3)) + (3 - 2) = 16 + 7 + 1 = 24.$$

Hence $m(6, 5, 25) \geq 40$.

Proposition 2. *Every 40-member family in $2^{[6]}$ has a 5-element trace of size at least 25.*

Proof. Suppose otherwise. By the standard squashing lemma [1, Lemma 1.1], the family may be replaced by a 40-member down-set with no larger traces. Denote this down-set by $\mathcal{F}$ and put $\mathcal{G} = 2^{[6]} \setminus \mathcal{F}$. Then $\mathcal{G}$ is an up-set of size 24. For $Y_x = [6] \setminus \{x\}$, the assumption and (2) imply

$$(4) \quad |\mathcal{G} \cap 2^{Y_x}| = 32 - |\mathcal{F}|_{Y_x} \geq 8 \quad (x \in [6]).$$

The family $\mathcal{G}$ contains no set of size at most one, since the upper cone of such a set has at least 32 members. It also contains no pair. Indeed, suppose $\{u, v\} \in \mathcal{G}$. Its upper cone has 16 members, so exactly eight members of $\mathcal{G}$ lie outside that cone. Equation (4) for $x = u$ forces all eight to avoid u , and the same equation for $x = v$ forces all eight to avoid v . If E is one of them, then $E \cup \{u\} \in \mathcal{G}$ because $\mathcal{G}$ is an up-set. This set still avoids v , so it lies outside the cone, but it contains u , a contradiction.

Thus every member of $\mathcal{G}$ has size at least three. For each x , equation (4) gives some member of $\mathcal{G}$ contained in Y_x ; the up-set property then gives $Y_x \in \mathcal{G}$. Hence all six 5-sets and $[6]$ belong to $\mathcal{G}$. Set

$$t = \left| \mathcal{G} \cap \binom{[6]}{3} \right|, \quad q = \left| \mathcal{G} \cap \binom{[6]}{4} \right|.$$

Then $t + q = 17$, and double counting pairs (x, G) with $G \in \mathcal{G} \cap 2^{Y_x}$ yields

$$48 \leq \sum_{x=1}^6 |\mathcal{G} \cap 2^{Y_x}| = 3t + 2q + 6.$$

Consequently, $t \geq 8$ and $q \leq 9$.

Choose eight triples from $\mathcal{G}$ and call the resulting family $\mathcal{H}$. The 4-element upper shadow of $\mathcal{H}$ is contained in $\mathcal{G} \cap \binom{[6]}{4}$. Under complementation in $[6]$, this upper shadow becomes the ordinary 2-element lower shadow of the eight triples $\{[6] \setminus H : H \in \mathcal{H}\}$. Since

$$8 = \binom{4}{3} + \binom{3}{2} + \binom{1}{1},$$

the Kruskal–Katona theorem [2, 3] gives

$$q \geq \binom{4}{2} + \binom{3}{1} + \binom{1}{0} = 10,$$

contrary to $q \leq 9$. □

The construction (3) and Proposition 2 prove $m(6, 5, 25) = 40$.

3. TWO COUNTEREXAMPLES ON SEVEN POINTS

Let

$$\mathcal{L} = \{123, 145, 167, 246, 257, 347, 356\}$$

be the lines of a Fano plane on $[7]$, and set

$$\mathcal{F}_4 = \binom{[7]}{\leq 3} \setminus \mathcal{L}.$$

Then $|\mathcal{F}_4| = 64 - 7 = 57$. Every 5-set is the complement of a pair. Exactly five Fano lines meet that pair, so exactly two Fano lines lie inside the 5-set. Therefore every 5-trace has size

$$\sum_{i=0}^2 \binom{5}{i} + \binom{5}{3} - 2 = 24,$$

and hence

$$(5) \quad m(7, 5, 25) \geq 58.$$

For the final construction, let

$$C = \{1, 4\}, \quad \mathcal{M} = \{123, 456, 147\},$$

and define

$$\mathcal{Q} = \left\{ Q \in \binom{[7]}{4} : Q \cap C \neq \emptyset \text{ and } M \not\subseteq Q \text{ for every } M \in \mathcal{M} \right\}.$$

There are $35 - \binom{5}{4} = 30$ four-sets meeting C . Each member of $\mathcal{M}$ has four four-set supersets, all meeting C , and these three upper cones are disjoint because the union of any two members of $\mathcal{M}$ has size at least five. Thus $|\mathcal{Q}| = 30 - 12 = 18$.

Now set

$$(6) \quad \mathcal{F}_5 = \binom{[7]}{\leq 2} \cup \left(\binom{[7]}{3} \setminus \mathcal{M} \right) \cup \mathcal{Q}.$$

By the definition of $\mathcal{Q}$, this is a down-set, and

$$|\mathcal{F}_5| = 29 + (35 - 3) + 18 = 79.$$

Let $Y_x = [7] \setminus \{x\}$. If $x \in C$, exactly one member of $\mathcal{M}$ lies in Y_x . Moreover, Y_x contains ten four-sets meeting C , exactly three of which contain that member of $\mathcal{M}$; hence $|\mathcal{Q} \cap 2^{Y_x}| = 7$. Therefore

$$|\mathcal{F}_5|_{Y_x} = 22 + (20 - 1) + 7 = 48.$$

If $x \notin C$, exactly two members of $\mathcal{M}$ lie in Y_x . Among the fifteen four-sets in Y_x , exactly one is disjoint from C . Each of the two contained members of $\mathcal{M}$ has three four-set supersets in Y_x , and no four-set contains both. Hence $|\mathcal{Q} \cap 2^{Y_x}| = 14 - 6 = 8$, and

$$|\mathcal{F}_5|_{Y_x} = 22 + (20 - 2) + 8 = 48.$$

Thus

$$(7) \quad m(7, 6, 49) \geq 80.$$

Equations (5) and (7) complete the proof of Theorem 1.

Theorem 1 disproves (1) as stated. It does not address whether the same formula might hold for all sufficiently large n when ℓ is fixed.

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